

When does the basic-SSA QP have no solution?

SPARK goal-insertion notes

The basic-SSA QP

The minimal-intervention safety filter solves, at each control step,

$$\min_u \frac{1}{2} \|u - u_{ref}\|_{Q_u}^2 \quad \text{s.t.} \quad \underbrace{L_g \phi_i u \leq -(L_f \phi_i + \eta)}_{\dot{\phi}_i \leq -\eta \text{ (safety)}}, \quad u_{\min} \leq u \leq u_{\max},$$

with $Q_u = \text{diag}(\text{control_weight})$. The goal does not appear in the objective; it enters only through the reference control u_{ref} .

The clean fact

The objective $\frac{1}{2} \|u - u_{ref}\|_{Q_u}^2$ is strictly convex ($Q_u \succ 0$) and bounded below by 0, so the QP is never unbounded and a minimizer exists if and only if the feasible set is nonempty. Hence

“no solution” $\iff$ the feasible region is empty.

The feasible region is the polytope

$$\mathcal{U} = \underbrace{\{u : u_{\min} \leq u \leq u_{\max}\}}_{\text{actuator box}} \cap \bigcap_{i \in \text{active}} \underbrace{\{u : L_g \phi_i \cdot u \leq -(L_f \phi_i + \eta)\}}_{\text{“make } \phi_i \text{ recede at rate } \geq \eta\}}.$$

Only the rows with $\phi_{i,\text{mask}_i} > 0 \wedge \phi_i \geq 0$ are active; eta_fn sets every other row's bound to $+\infty$. Infeasibility means this intersection is empty.

The three ways it goes empty

(A) Conflicting active constraints — the dominant cause

Two or more body–obstacle pairs are active at once and the retreat directions they demand are mutually incompatible: no single u makes all active safety indices decrease at rate η simultaneously. The arm is pincered. With $A = L_g \phi$ and $b = -(L_f \phi + \eta)$ over the active rows, Farkas/Gordan gives

$$\{u : Au \leq b\} = \emptyset \iff \exists y \geq 0 : A^\top y = 0 \text{ and } b^\top y < 0,$$

i.e. a nonnegative blend of the constraint normals cancels out while the offsets stay jointly negative — “obstacle A pushes me $+x$, obstacle B pushes me $-x$, and I must retreat from both at once.” This is exactly the pinch the inserted goal G'_1 engineers.

(B) Actuator saturation vs. the required rate

Even a single constraint is infeasible if the demanded braking exceeds what the bounded control can produce — the most-favorable corner of the box still violates the bound:

$$\min_{u \in [u_{\min}, u_{\max}]} L_g \phi_i \cdot u > -(L_f \phi_i + \eta).$$

A larger η (steeper required descent) or a large positive $L_f \phi_i$ (fast uncontrolled closing) tightens this. For single-integrator dynamics $\dot{x} = u$ the drift $L_f \phi = 0$, so this reduces to “ η is larger than the box allows along the $-\nabla \phi$ direction” — mild alone, but it shrinks $\mathcal{U}$ and makes (A) easier to trigger.

(C) Degenerate control authority

An active constraint with $L_g \phi_i = 0$ (the control cannot influence that body-pair’s distance at this configuration — a singular Jacobian, or the index gradient orthogonal to every controllable direction) while $\phi_i \geq 0$ demands a decrease. That row becomes $0 \cdot u \leq b_i$ with $b_i = -\eta < 0$, i.e. $0 \leq -\eta$, which is impossible.

In our setting specifically

Because we run single-integrator dynamics ($L_f \phi = 0$) with a modest η , infeasibility is almost entirely (A) — conflicting simultaneous constraints — narrowed by (B)’s box. At that pinch:

- basic SSA returns infeasible $\rightarrow$ falls back to $u_{ref} \rightarrow$ unfiltered motion $\rightarrow$ collision (and the infeas counter ticks);
- r-SSA / p-SSA never reach this state — they add a slack s , relaxing the bound to $-L_f \phi + s$, so their feasible set is always nonempty by construction. They trade the infeasibility for either large slack (push-through collision under low penalty, p-SSA) or a frozen min-slack equilibrium (deadlock under high penalty, r-SSA).

Same pinch, three different failure expressions.

One-line answer

The basic-SSA QP has no solution exactly when the actuator box and the active safety half-spaces have empty intersection — overwhelmingly because the active obstacle constraints conflict (a

pincer), secondarily because actuator limits cannot meet the required braking rate. That empty-intersection event is the geometric object a coverage-guided fuzzer (the pinch-configuration descriptor) would map.